



Securing the internet of things: A comprehensive review of ransomware attacks, detection, countermeasures, and future prospects

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ABSTRACT

Ransomware attacks present a critical and persistent threat by denying access to data until a ransom is paid. Although existing research on IoT ransomware often concentrates on specific attack types or industry sectors, it frequently overlooks newly emerging threats and lacks holistic defense frameworks applicable across diverse IoT ecosystems. To bridge these gaps, this paper provides a comprehensive analysis of the evolving ransomware landscape targeting IoT devices. We systematically review and categorize detection technologies—ranging from signature-based methods to cutting-edge AI-driven solutions—and assess their effectiveness in mitigating these threats. Additionally, we propose a multi-layered defensive strategy integrating technological, legal, and policy measures to address the complexities of ransomware in IoT settings. Looking ahead, our study highlights potential research directions such as advancing real-time detection, leveraging blockchain for enhanced security and fostering cross-sector collaboration to bolster collective threat intelligence. By emphasizing the importance of a unified approach that involves researchers, industry professionals, and policymakers, we underline the critical steps necessary to fortify IoT infrastructures against the ever-evolving ransomware threat.

1. Introduction

The growing prevalence of IoT devices in daily life and critical infrastructure introduces significant security challenges [1]. Ransomware attacks, which deny users access to their data until a ransom is paid, have become increasingly sophisticated, posing serious risks to individuals and organizations [2]. In 2023, Europe faced 55 ransomware operations affecting 906 victims, with the Lockbit operation accounting for 25 % of cases [3]. Similarly, the United States experienced a significant rise in ransomware incidents, particularly in healthcare, where 141 hospitals were impacted, highlighting vulnerabilities in critical infrastructure [4].

IoT devices often lack robust security features, making them susceptible to attacks [5]. Reports indicate that in 2022, the industry suffered at least 437 ransomware attacks, marking a 107 % increase from the previous year. These attacks often target operational technology, leading to production halts and substantial financial losses. Studying ransomware in IoT is critical, as existing research often lacks a holistic perspective on diverse threats and emerging contexts [6]. While historical analyses provide useful insights, advanced and adaptable security strategies are urgently needed to address the unique vulnerabilities of IoT ecosystems.

Municipal governments face similar vulnerabilities due to their reliance on interconnected systems for public safety communications, water supply management, and transportation networks. Ransomware attacks can impair emergency response coordination, disrupt water supplies, and compromise traffic systems. For example, a city may be infected after an employee opens a phishing email, resulting in demands of \$500,000 or more. These scenarios highlight the critical need for robust cybersecurity strategies, including regular updates, employee training, and effective incident response plans [7].

The financial sector, with its sensitive data and potential for high payouts, has also become a prime target. Financial institutions face operational disruptions and substantial costs to secure systems and mitigate threats. Educational institutions and government agencies, which store vast amounts of sensitive information, are increasingly targeted, raising concerns about data privacy and national security [8]. As the impacts of ransomware extend across critical sectors, another vulnerable area emerges in the form of connected technologies.

A ransomware attack occurs when malicious software encrypts a victim's files or locks them out of their system, demanding payment to restore access. The attack often begins when the victim unknowingly opens a malicious email attachment, clicks on a compromised link, or encounters a vulnerability in their system [9]. Once inside, the

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ransomware spreads, encrypts files, and disables security features. The attacker then delivers a ransom note, typically demanding payment in cryptocurrency to maintain anonymity, often with a deadline to pressure the victim. In some cases, attackers may threaten to leak sensitive data if the ransom is not paid. These attacks cause operational disruption, financial loss, and reputational damage, making them a significant cybersecurity threat [10]. Fig. 1 overviews the detailed visualization of the how a ransomware attack can progress, as discussed earlier.

This study aims to fill these gaps by thoroughly analyzing the impact of ransomware on IoT systems and proposing new strategies for detection and mitigation. It combines current knowledge, identifies the limitations of existing methods, and suggests integrated solutions using advanced technologies like machine learning and blockchain. The contributions of this study are multidimensional. It provides an in-depth examination of how ransomware uniquely affects IoT devices, bridging the gap between general ransomware research and the specific challenges of IoT. Furthermore, it proposes innovative defense mechanisms specifically designed for the IoT environment, improving both theoretical understanding and practical applications in addressing these threats. By detailing the vulnerabilities of IoT devices to ransomware attacks, this study aims to enhance public and corporate awareness of this issue. Moreover, it emphasizes the urgent need for more flexible and advanced defense mechanisms. Finally, the findings of this study are expected to provide an empirical basis for relevant legal and policy development, promoting effective regulations and policy innovations to address increasingly complex cybersecurity challenges. This study underscores the importance of tailored strategies that can be effectively implemented in various IoT contexts, such as smart cities and industrial systems. By doing so, it aims to guide future research and offer practical solutions to protect the increasingly connected world of IoT from the rising threat of ransomware attacks. To summarize, the main key of this survey is defined as follows:

- Providing a comprehensive overview of evolution of ransomware attacks on IOT,
- Outlining a classification of various ransomware attacks targeting IOT,
- Discussing a brief classification of detection and countermeasure techniques,
- Highlighting current challenges and potential future directions associated with ransomware attacks on IOT.

The remainder of this paper is organized as follows: Section II describes related works. Section III illustrates the basic facts of Ransomware Attacks. Section IV makes a classification of ransomware attacks.

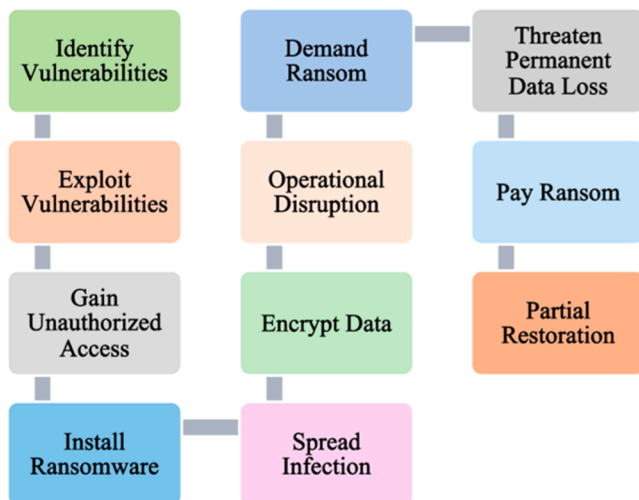


Fig. 1. Visualization of how ransomware attack progress.

Section V introduces different detection techniques, while Section VI provides an overview of countermeasure techniques. In Section VII, future directions and challenges are addressed. Finally, Section VIII highlights the conclusion.

2. Related works

There are several studies have explored the significance of ransomware attacks, as summarized in Table 1. These articles have been categorized into three distinct themes. The first section examines detection techniques for ransomware attacks and IoT botnets, highlighting key advancements in machine learning and deep learning approaches. For instance, [11] reviewed crypto-ransomware detection methods, emphasizing prevalent attack types and adaptive security measures. The study identifies critical challenges and proposes future research directions. Similarly, [12] introduced MalBoT-DRL, a detection system utilizing Deep Reinforcement Learning to identify IoT botnets across their lifecycle. It achieved up to 99.94 % accuracy on the MedBioT and N-BaIoT datasets with minimal computational overhead. Furthermore, [13] leveraged Vision Transformers (ViT) to enhance intrusion detection in IoT networks, demonstrating superior performance over traditional models in detecting complex network anomalies using the NSL-KDD dataset. Another study, [14], proposed SecurityBERT, a lightweight BERT-based model capable of achieving 98.2 % accuracy in detecting 14 IoT attack types. The system is optimized for resource-constrained environments, offering efficiency and privacy preservation.

The second section explores the impacts of ransomware in critical sectors such as healthcare and smart cities, along with mitigation strategies aimed at enhancing cybersecurity. The increasing prevalence of ransomware in critical domains, including healthcare and smart cities, has been extensively studied. For instance, [15] assessed the impact of ransomware on the healthcare sector, highlighting the heightened risks posed by digitalization and the significant value of medical data. The study outlines mitigation strategies, such as blockchain and machine learning, while addressing challenges in implementing effective defenses. Similarly, [16] analyzed the risks of ransomware in healthcare, categorizing solutions like blockchain, software-defined networking (SDN), and machine learning, and identifying barriers to creating robust countermeasures [17]. explored the implications of ransomware on IoT systems within smart cities and critical infrastructure, emphasizing the disruptions and financial losses caused. This study advocates for employing AI, blockchain, and SDN to develop innovative security solutions.

The third section reviews the evolution of ransomware from its inception to advanced encryption techniques, along with the development of state-of-the-art detection models. Several studies have examined the evolution of ransomware and the development of advanced detection models [18], provided a comprehensive review of ransomware's progression from 1989 to 2020, documenting its evolution from simple locking mechanisms to advanced encryption methods such as AES and RSA. The study underscores the increasing use of phishing and cryptocurrency-based payment systems and highlights the importance of updated security measures. In terms of advanced detection models, [19] introduced iCNN-LSTM, which integrates CNNs (Convolutional Neural Network), LSTMs (Long short-term memory), and attention mechanisms to analyze Sysmon logs. This model employs batch-based incremental learning for real-time adaptation to new ransomware variants, achieving a 99.61 % F2-score, with low false-positive and false-negative rates, making it highly suitable for imbalanced datasets. Similarly, [20] investigated ransomware threats in operational technologies, emphasizing the necessity of machine learning for effective detection and the development of sophisticated defense mechanisms.

This research seeks to build upon existing knowledge while addressing the identified gaps. Specifically, it aims to enhance our understanding of the impact of ransomware on IoT systems by examining both the technical implications and the broader operational challenges.

Table 1
Summary of related works.

References	Year	Area of focus	What is covered	What is not covered?
[11]	2022	Crypto Ransomware Attacks	State-of-the-art methodologies for detecting attacks Detection models Evasive mechanisms	Specific case studies or examples of attacks
[12]	2023	IoT and Ransomware	intrusion detection Deep reinforcement learning	Reliance on specific datasets Challenges with Early-Stage Detection
[13]	2021	IoT and Ransomware	Vision transformers IoT Deep learning	Uses the NSL-KDD dataset, which may not capture modern ransomware complexities.
[14]	2024	IoT and Ransomware	Ransomware threat detection IoT networks Generative AI	Privacy-preserving techniques may introduce computational overhead for real-time ransomware response.
[15]	2021	Ransomware in Healthcare Systems	Vulnerabilities in healthcare Mitigation strategies Blockchain and Machine Learning	Specific healthcare data protection regulations
[16]	2021	Ransomware in Healthcare Systems	Risks and vulnerabilities of healthcare systems to ransomware Blockchain, SDN technology, and Machine Learning	Privacy-preserving techniques may introduce computational overhead for real-time ransomware response.
[17]	2022	Ransomware in Smart Cities and Infrastructure	Use of AI, Blockchain, and SDN against ransomware in smart cities	Specific smart city technologies.
[18]	2024	Advanced IT Security Systems	Ransomware evolution, use of AES and RSA encryption Adaptation of security measures	Specific applications in IoT, Healthcare, Smart Grid, or Drones
[19]	2025	IoT and Ransomware	Ransomware detection Deep Learning Incremental learning	Relies heavily on Sysmon logs Scalability for larger data streams is not evaluated
[20]	2024	Ransomware Detection Techniques	Detection strategies Use of machine learning for ransomware detection Need for innovative solutions	Long-term effectiveness of these technologies.

To achieve this, we propose an integrated strategy that harnesses recent advancements in machine learning, blockchain technology, and comprehensive security protocols. This approach is designed to deliver a more robust and resilient defense against ransomware threats.

This chapter reviewed recent literature on ransomware detection and

mitigation, revealing gaps in scalability, applicability, and cross-disciplinary collaboration. Building on these findings, the next chapter will delve into ransomware attacks more closely, examining their key characteristics, propagation methods, and impacts on IoT systems.

3. Overview of ransomware attacks

Ransomware attacks involve malicious software designed to block access to computer systems or data until a ransom is paid. These attacks often begin with a user unknowingly clicking on a malicious link or email attachment, which installs ransomware on their device. Once activated, the ransomware encrypts files, rendering them inaccessible. Attackers then demand payment, often in cryptocurrencies like Bitcoin, in exchange for a decryption key [19]. Ransomware attacks are not only disruptive but also have widespread implications across various domains.

Ransomware has severe implications for critical sectors, including healthcare, finance, and government services. In healthcare, ransomware can encrypt vital patient records, disrupt medical devices, and incapacitate scheduling systems. These disruptions delay treatments and may force hospitals to divert emergency patients, endangering lives [20]. Attackers often demand significant ransoms, such as \$1 million in Bitcoin, leaving organizations with the difficult choice of paying the ransom or restoring systems from backups, which is time-consuming and costly [21].

The rapid adoption of Internet of Things (IoT) devices has introduced new vulnerabilities. IoT devices, ranging from smart home appliances to industrial control systems, often lack robust security features, making them attractive targets for ransomware. Compromised IoT devices can disrupt essential operations in sectors like healthcare and manufacturing. For instance, infected medical devices can jeopardize patient safety, while industrial IoT disruptions can lead to production downtime and safety hazards [22]. The interconnected nature of IoT ecosystems amplifies the impact of a single compromised device, complicating recovery efforts [23]. Ensuring IoT security requires coordinated efforts from manufacturers, policymakers, and cybersecurity experts to establish and enforce robust security standards. These measures are critical to protecting IoT devices from ransomware threats and maintaining user trust in emerging technologies [24]. With vulnerabilities spanning critical infrastructure and IoT devices, the global trends in ransomware incidents further illustrate the severity of this issue.

Ransomware attacks have increased dramatically worldwide, with the United States being one of the most affected countries. According to the SANS Institute, ransomware incidents in the U.S. rose by 73 % from 2022 to 2023, with 4611 cases reported in 2023 compared to 2662 in 2022 [25]. High-profile attacks on hospitals, financial institutions, and educational institutions underscore the pervasive threat ransomware poses across sectors. Other countries, including the United Kingdom, Germany, and Canada, have reported similar trends, highlighting the global nature of the threat.

To understand the scope of the growing ransomware threat, it is crucial to examine its historical evolution and the increasingly sophisticated methods employed by attackers. Ransomware has evolved significantly since its inception in the late 1980s. One of the earliest examples is the AIDS Trojan (also known as Aids Info Disk or PC Cyborg Trojan), which spread via floppy disks and demanded payments sent through postal mail [26]. Created by Joseph Popp, this malware was a primitive yet pioneering form of cyber extortion, forcing victims to pay \$189 to a post office box in Panama to unlock their files [27]. As technology advanced, ransomware became increasingly sophisticated. By 2005, Gpcode utilized RSA encryption, requiring a unique decryption key for each victim, which significantly complicated recovery efforts without payment [28]. The 2013 CryptoLocker attack marked another milestone, employing Bitcoin for ransom payments and strong encryption methods. This attack demonstrated the lucrative potential of ransomware, impacting over 250,000 systems worldwide [29].

A turning point in ransomware evolution came in 2017 with the WannaCry and NotPetya attacks. WannaCry exploited a Windows vulnerability via the NSA-developed EternalBlue exploit, affecting numerous systems globally, including the UK’s National Health Service, and disrupting medical services. Conversely, NotPetya primarily aimed to destroy data rather than collect ransoms. Initially disguised as ransomware, it targeted Ukrainian infrastructure but caused widespread damage to global corporations, resulting in billions of dollars in losses [30,31].

In recent years, ransomware threats have escalated further. Since 2018, sophisticated ransomware strains like Ryuk have targeted high-value sectors, including healthcare and government. Ryuk’s approach involves disabling backups and security tools, making recovery nearly impossible without paying substantial ransoms [32,33]. Other dangerous variants, such as Maze, Sodinokibi/REvil, and DoppelPaymer, emerged in the late 2010s, reinforcing the urgency for robust cybersecurity measures [34]. The 2020–2021 period saw the rise of new groups like DarkSide, infamous for the Colonial Pipeline attack, which disrupted U.S. fuel supplies significantly [35]. These groups introduced "double extortion" tactics, combining data encryption with data theft to increase pressure on victims. By 2021 and 2022, ransomware strains like BlackCat/ALPHV [36], Lorenz [37], and Grief [38], alongside the Conti group [39], adopted even more advanced distribution methods, including phishing and exploit kits. High-profile cases, such as those involving Nvidia and SpiceJet, underscored the evolving threat posed by ransomware [40]. By 2023, the emergence of the 3CX Supply Chain Attack marked a new tactic, focusing on supply chain infiltration to maximize impact and ransom potential [41]. This approach highlights the interconnected nature of modern digital ecosystems and the extensive consequences an attack on one provider can have across an entire network [42].

Table 2 summarizes the evolution of ransomware attacks, detailing their characteristics, strengths, limitations, and representative examples. The historical progression from the late 1980s to the present day illustrates the continuous adaptation of ransomware tactics to exploit vulnerabilities in increasingly complex systems. This trend underscores the critical need for accelerated research into advanced threat detection systems and comprehensive employee training programs. These measures are essential to mitigating risks and protecting sensitive data from sophisticated ransomware threats.

In this chapter, we highlighted how ransomware disrupts different industries by exploiting vulnerabilities in IoT devices. Understanding these diverse attack scenarios is crucial for devising effective responses. The following chapter will classify ransomware types and illustrate their distinct threat patterns in IoT contexts.

4. Classification of ransomware attacks

Ransomware attacks targeting IoT devices exploit several unique characteristics that distinguish them from attacks on traditional IT systems. First, the diversity and ubiquity of IoT devices present a broad and varied attack surface. IoT devices range from everyday consumer products, such as smart refrigerators and home security cameras, to critical infrastructure components, including industrial control systems and medical devices. This diversity embeds IoT devices into nearly every aspect of modern life, often without consistent or comprehensive security measures [43]. Their widespread deployment across various environments—from homes and offices to factories and hospitals—further complicates efforts to secure them against cyber threats.

Many IoT devices are designed with limited processing power and memory, significantly hampering the implementation of robust security mechanisms [44]. Unlike traditional computing systems capable of running comprehensive antivirus and anti-malware programs, IoT devices often operate on minimal resources, making them incapable of supporting such protective software. Moreover, these devices frequently rely on specialized or outdated operating systems that do not receive

Table 2
Overview of the evolution of ransomware attacks [26–42].

Ransomware	Year	Characteristics	Strengths	Limitations
AIDS Trojan	1989	Spread via floppy disks Demanded payment via postal mail	Pioneered the concept of digital extortion	Primitive method Low tech Easily countered
Gpcode	2004	Used RSA encryption Required a unique decryption key	Advanced encryption increased difficulty of data recovery without payment	Encryption could be broken with sufficient computing power
CryptoLocker	2013	Used Bitcoin for ransom Strong encryption	Utilized cryptocurrencies for anonymity High profit from widespread impact	Vulnerable to law enforcement and security research takedowns
WannaCry	2017	Exploited Windows vulnerability with EternalBlue	Rapid global spread Caused massive disruption Especially in healthcare	Patch for vulnerability was available before the attack, limiting scope
NotPetya	2017	Masqueraded as ransomware but primarily aimed to destroy data	Caused extensive damage to multinational corporations	Not aimed at revenue
Ryuk	2018	Targets high-value sectors uses advanced malware for entry	Disables backups Making recovery without payment difficult	More destructive Requires initial foothold Can be prevented
Maze	2019	Pioneered double extortion tactics	Stole and encrypted data Increase pressure to pay	Dependent on initial access via phishing or other vulnerabilities
Sodinokibi/REvil	2019	Known for auctioning stolen data if ransoms aren't paid	Highly profitable through aggressive extortion tactics	High public and law enforcement attention can lead to shutdowns
DoppelPaymer	2019	Known for attacking public agencies and demanding high ransoms	Targeted attacks could result in large payouts	Vulnerable to cybersecurity defenses if properly implemented
DarkSide	2021	Responsible for the Colonial Pipeline attack	Sophisticated operational capabilities Targeted critical infrastructure	Operations can be ceased under government pressure
BlackCat/ALPH	2021	Uses sophisticated ransomware-as-a-service model	High customization of attacks targets a wide range of industries	Relies on affiliates for distribution, which can be a weak link
3CX Supply Chain Attack	2023	Targets supply chains to maximize impact	Exploits interconnected systems for widespread disruption	Requires sophisticated planning and execution High risk of detection

regular security updates. This lack of updates leaves IoT devices particularly vulnerable to ransomware attackers, who can exploit known vulnerabilities to gain control [45].

Another critical vulnerability is the inherent connectivity of IoT devices. Designed to remain continuously connected to other devices and central management systems via various networks, including the internet, this interconnectedness is essential for their functionality but

also introduces significant security risks [46]. A single compromised device can act as an entry point for attackers to infiltrate and compromise other connected devices within the same network. This facilitates the rapid spread of ransomware across multiple devices, causing widespread disruption and complicating containment efforts.

To effectively counter these threats, it is essential to understand the various types of ransomware-based attacks targeting IoT devices. These attacks can be classified into three categories: crypto-based, locker-based, and hybrid-based attacks, as illustrated in Fig. 2. The following discussion provides a brief overview of each attack type, detailing their functionality, impacts, strengths, and limitations. Table 3 summarizes the key features of these attack types, including their year of emergence, unique characteristics, use cases, advantages, and disadvantages:

4.1. Crypto-based ransomware attacks

Crypto ransomware specifically tailored for IoT environments presents a significant threat by infiltrating devices and encrypting vital files such as configuration data, logs, and personal information, rendering them inaccessible without a decryption key [47]. These attacks utilize advanced encryption algorithms to effectively lock down these files, after which a ransom note is displayed, providing instructions for payment—typically in cryptocurrency to preserve the attacker’s anonymity [48]. The consequences for victims can be severe, as essential device functionalities are disabled, leading to substantial disruptions across personal, commercial, and industrial domains [49]. Victims are often faced with a difficult decision: lose access to critical data or comply with ransom demands, thereby indirectly supporting criminal activities.

For instance, in 2019, a surge of crypto ransomware attacks targeted smart energy meters across Europe. Exploiting a firmware vulnerability, the attackers encrypted essential operational data, resulting in widespread disruptions to monitoring and billing systems [50]. Utility companies not only experienced immediate service interruptions but also faced the long-term challenge of rebuilding customer trust. In some

cases, these companies opted to pay the ransom to expedite service restoration, underscoring the urgent need for robust security measures in IoT ecosystems [51]. Fig. 3 illustrates the mechanisms through which crypto-based ransomware attacks are executed.

Several examples of crypto-based ransomware attacks are depicted in Fig. 2. In 2017, the Hajime ransomware, renowned for its adaptability, encrypted data on a wide range of IoT devices, including routers and security cameras, affecting thousands of systems worldwide. This resulted in widespread service disruptions and exposed significant security vulnerabilities. Affected organizations and individuals were compelled to either restore systems from backups or comply with ransom demands, highlighting the evolving threat landscape posed by such adaptable malware [52]. Also in 2017, BrickerBot targeted poorly secured IoT devices, such as home routers, causing irreparable damage by corrupting their storage systems. This rendered the devices completely inoperable, necessitating their replacement and underscoring the urgent need for improved security practices in IoT device management [53,54]. In 2018, a variant of Gafgyt infiltrated IoT devices within a healthcare facility, encrypting critical files and disrupting operations. This attack delayed patient care, led to significant financial losses, and emphasized the healthcare sector’s reliance on robust cybersecurity measures to safeguard its digital infrastructure against sophisticated threats [55].

In 2019, Echobot primarily targeted IoT devices within an industrial plant. By exploiting system vulnerabilities, the attackers encrypted critical operational data, severely disrupting manufacturing processes and forcing temporary plant shutdowns. The attack caused substantial financial losses and necessitated negotiations with the attackers to obtain decryption keys, further underscoring the importance of robust security measures for industrial IoT environments [56]. Finally, in 2020, IoTReaper combined characteristics of earlier ransomware strains with enhanced capabilities to target smart home ecosystems. By encrypting home automation devices, the malware effectively locked homeowners out of essential systems, from lighting controls to security measures. This attack caused significant personal inconvenience and exposed critical vulnerabilities in smart home networks.

In summary, crypto ransomware tailored for IoT environments operates by infiltrating devices and encrypting sensitive files, such as configuration data, logs, and personal information, rendering them inaccessible without a decryption key. Employing advanced encryption algorithms, these attacks secure the files and deliver a ransom note instructing victims to pay in cryptocurrency, ensuring the anonymity of the attackers. Such attacks result in catastrophic scenarios where critical device functions are disabled, causing substantial disruptions across personal, commercial, and industrial domains. Victims are forced to choose between losing vital data or funding criminal activities. The effectiveness of these attacks lies in the sophistication of the encryption, which is often nearly impossible to bypass without the appropriate decryption key. The implications are severe, as ransomware can bring entire operations to a halt, emphasizing the need for proactive and robust cybersecurity measures.

4.2. Locker-based ransomware attacks

Locker ransomware, distinct from its counterpart crypto ransomware, disables access to IoT devices without encrypting the files stored on them [57]. This type of ransomware focuses on seizing control of the entire system or critical components, such as user interfaces and control panels, effectively locking the user out. Once the device is locked, a ransom note is prominently displayed on the device’s interface. This note typically contains instructions for payment—often in cryptocurrency—to regain access. By invoking urgency and fear, attackers aim to coerce victims into complying quickly to restore device functionality. Unlike the complex encryption methods used in crypto ransomware, locker ransomware relies on its ability to disrupt access, turning every locked device into a potential revenue stream for attackers. The

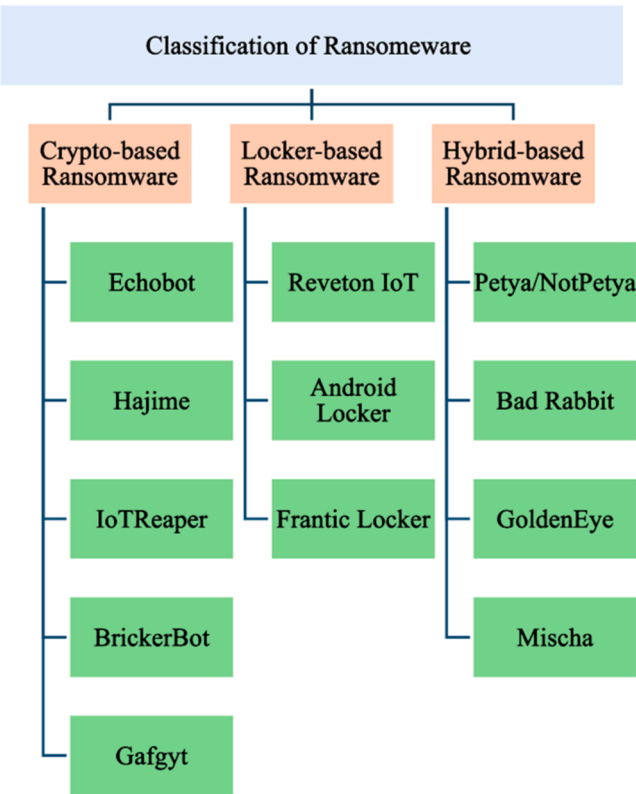


Fig. 2. Classification of ransomware-based attacks on IoT.

Table 3

Overview of three types of ransomware attacks.

Type	Year	Key Features	Use Cases	Advantages	Disadvantages
Crypto-based Ransomware	1989	Encrypts crucial files making them inaccessible without a decryption key Uses strong encryption algorithms	IoT environments in various sectors	High security through encryption Can halt operations leading to higher chances of ransom payment Rapidly disables access Simplifying the attack process.	Recovery is difficult without decryption keys Depend on victims not being able to bypass encryption Can often be bypassed without paying ransom if backups or exploits are available.
Locker-based Ransomware	2007	Locks access to devices without encrypting files Seizes control of critical system components	Consumer electronics Public kiosks	Doubles recovery challenges Enhance effectiveness by using both data loss and operational disruption	Complex, potentially leaving more traces or flaws for cybersecurity professionals to exploit.
Hybrid-based Ransomware	2012	Combines tactics of crypto and locker ransomware encrypts files and locks system access	High-value targets across all sectors		



Fig. 3. Process of a crypto-based ransomware attack.

immediate and disruptive nature of these attacks highlights the dependence on uninterrupted access to digital systems in modern personal and business environments [58].

An illustrative case occurred in 2016, when a popular model of LG smart TVs was targeted by locker ransomware. The attack locked the screen and displayed a ransom note, falsely claiming to be from a federal agency and accusing users of engaging in illegal activities. The note demanded payment in cryptocurrency to unlock the device. Many users, unable to access their TVs, chose to pay the ransom. However, some were able to restore functionality by performing a factory reset as advised by technical support [59]. Fig. 4 illustrates the operational mechanisms of locker-based ransomware attacks.

As depicted in Table 3, several instances of locker-based ransomware attacks have been recorded, highlighting the evolving threat landscape. A notable example is the Reveton IoT ransomware, which in 2017 compromised public information kiosks. The ransomware displayed a fraudulent note accusing users of accessing illegal content and demanded payment of fines. This rendered the kiosks inoperable until the ransom was paid or the systems were restored, emphasizing the vulnerability of public devices to locker ransomware attacks. This incident underscored the urgent need for enhanced security measures for devices that serve the public [60].

Another significant case occurred in 2018, when the City of Atlanta, Georgia, experienced a major disruption due to a SamSam ransomware attack. Although SamSam primarily utilized encryption rather than locker tactics, its impact on public services warrants attention. The attack exploited a server-side software vulnerability, encrypting critical

systems and affecting 8000 city employees. Residents were unable to access essential services such as utility payments and the parking ticket system. This incident revealed critical security weaknesses in public infrastructure systems and highlighted the necessity for robust cybersecurity measures in government networks to mitigate such risks [61].

In March 2019, LockerGoga ransomware paralyzed the operations of Norsk Hydro, a major Norwegian aluminum producer. The ransomware infiltrated the company's operational systems through spear phishing, locking critical data and forcing several plants to operate manually. Unlike traditional ransomware that demands payment for data recovery, LockerGoga aimed to disrupt operations entirely, representing an evolution in cyber threat strategies targeting both digital and physical areas of a company. This incident further emphasized the vulnerability of industrial environments to sophisticated cyber threats and underscored the importance of robust cybersecurity frameworks and regular system updates to safeguard against operational and digital disruptions [62].

4.3. Hybrid-based ransomware attacks

Hybrid ransomware represents a significant security threat, as it combines the tactics of both crypto and locker ransomware to intensify its impact. This malware not only encrypts critical files but also locks users out of their systems, creating a dual challenge that complicates recovery efforts and heightens the urgency to comply with ransom demands. By simultaneously encrypting vital data and restricting access to device interfaces, hybrid ransomware severely limits both operational capabilities and data retrieval options [63].

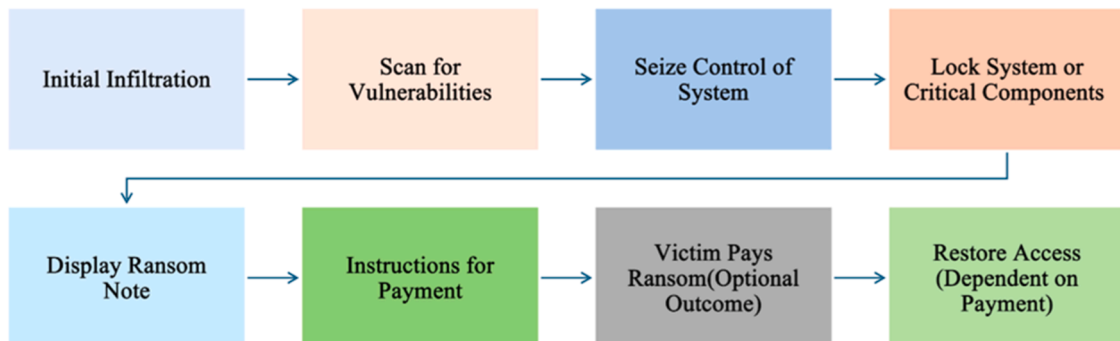


Fig. 4. Process of a locker-based ransomware attack.

To further pressure victims, it typically delivers a ransom note with detailed instructions for both decryption and system unlocking, often demanding a higher ransom due to the compounded attack approach. The dual-function strategy employed by hybrid ransomware is particularly effective at coercing victims, resulting in severe disruptions and an increased risk of substantial data loss [64]. For example, [65] underscores that hybrid ransomware can disrupt operations by combining encryption with lockdown techniques, effectively paralyzing infrastructure and exposing organizations to significant security risks. This highlights the critical importance of implementing robust cybersecurity measures to defend against such sophisticated and multifaceted threats. Fig. 5 illustrates the operational mechanism of hybrid ransomware attacks.

Table 3 presents a series of hybrid ransomware attacks that illustrate the complexity and widespread impact of this dual-function threat. A prominent example is Petya/NotPetya, which emerged in 2017 and initially targeted businesses in Ukraine by encrypting the Master Boot Record (MBR) of affected computers. The attack quickly escalated into a global crisis, causing unprecedented operational disruptions and inflicting billions of dollars in damage across multiple sectors, including healthcare, finance, and logistics. This incident highlighted the devastating consequences of sophisticated hybrid ransomware attacks and underscored the critical need for robust cybersecurity measures across all industries [66].

Another notable example is Bad Rabbit, which surfaced later in 2017, cleverly disguised as Adobe Flash updates. The ransomware spread rapidly across Eastern Europe, compromising media outlets and critical infrastructure by encrypting files and locking systems. This attack demonstrated the effectiveness of social engineering tactics in spreading ransomware, severely disrupting critical services and emphasizing the importance of maintaining strict digital hygiene and vigilant update management practices [67].

GoldenEye ransomware also gained attention in 2016, disrupting the operations of a multinational corporation by encrypting both employee files and the MBR. The attack caused significant operational downtime and extensive data losses, serving as a stark reminder of the destructive potential of hybrid ransomware. This event prompted many organizations to reevaluate their cybersecurity strategies, stressing the importance of multi-layered, up-to-date security solutions to combat evolving cyber threats [68].

In summary, hybrid ransomware poses a particularly formidable threat due to its combination of crypto-ransomware and locker ransomware tactics, amplifying the severity of its impact. These attacks simultaneously encrypt critical data and lock users out of their systems, doubling the challenges of recovery and increasing the urgency to comply with ransom demands. By exploiting both data loss and operational disruption, hybrid ransomware exerts significant pressure on victims. While this complexity makes it a powerful tool for attackers, it

can also leave behind additional traces and vulnerabilities for cybersecurity professionals to identify and exploit.

Having categorized crypto-based, locker-based, and hybrid ransomware attacks, we can now better understand how each type targets IoT systems. The next chapter examines various detection techniques—ranging from signature-based to AI-driven approaches—and assesses their effectiveness against these evolving threats.

5. Detection techniques

In this section, we present a classification of ransomware detection techniques for IoT environments, followed by a comprehensive discussion of these techniques and a comparison based on various characteristics.

5.1. Classification

Detection and classification techniques for ransomware attacks on IoT devices can be categorized into eleven distinct groups, as illustrated in Fig. 6. These categories include Signature-Based, Heuristic-Based, Behavioral Analysis-Based, Anomaly Detection-Based, Artificial Intelligence-Based, Deception-Based, Sandboxing-Based, Real-Time Protection Systems, File Integrity Monitoring-Based, Cloud-Based, Blockchain-Based, and Hybrid-Based techniques. Each of these detection approaches is briefly discussed below:

5.1.1. Signature-based detection

Signature-based detection operates through a series of technical processes designed to identify known ransomware threats with precision and efficiency [69]. At its core, this method uses signatures, which are digital representations of unique features extracted from previously identified ransomware samples. These features may include file hashes computed using algorithms like MD5 or SHA256, unique sequences of assembly instructions, or specific API call patterns, by analyzing the frequency and order of these interactions, systems can generate dynamic signatures that effectively differentiate between malicious and benign applications. For example, Locky ransomware was effectively detected using signature-based methods that identified its encryption algorithm patterns. These signatures are stored in a database, enabling real-time comparison with incoming files or network traffic to detect potential threats [70].

The detection workflow begins with a system scan of files, memory, or network packets, applying pattern-matching algorithms to identify matches with the stored signatures. Hash-based detection is particularly effective for identifying exact replicas of malicious files, as even a single byte change in the file content results in a completely different hash. Beyond hash comparisons, some systems analyze code structures or embedded scripts to detect ransomware variants that might exhibit the

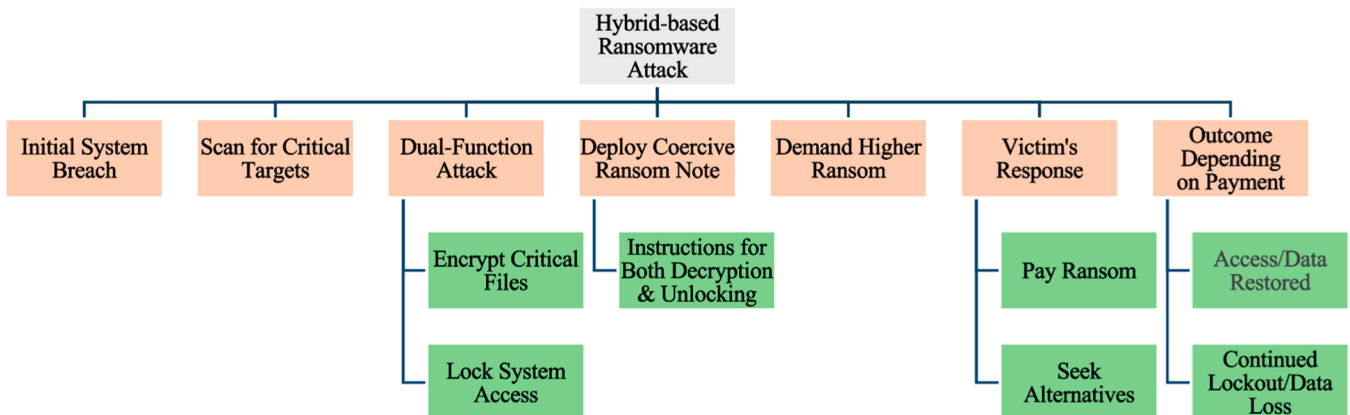


Fig. 5. Process of a hybrid-based ransomware attack.

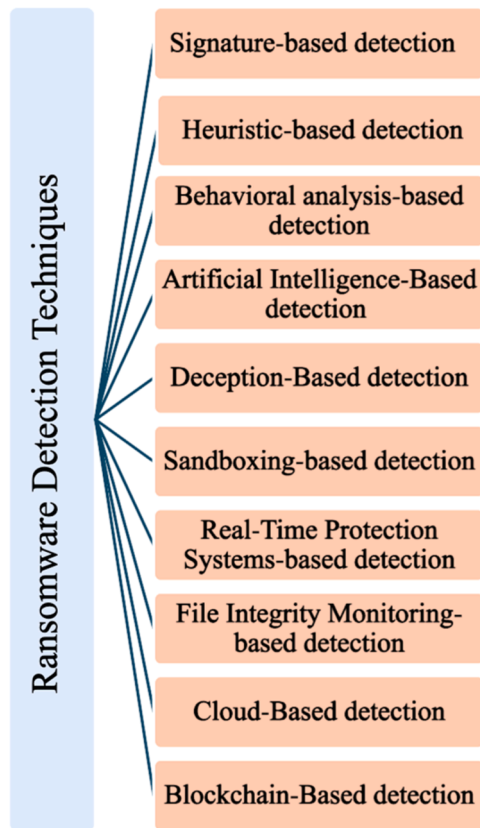


Fig. 6. Classification of ransomware detection techniques on IoT.

same functional logic, even if their binary representation has been altered.

Behavioral signature-based detection extends the technique by encoding dynamic traits, such as encryption routines, frequent access to sensitive file paths, or irregular communication with command-and-control servers. These behavioral patterns, once observed and documented in known ransomware, are transformed into signatures that help in recognizing new attacks exhibiting similar behavior [71].

Another advancement is the development of preemptive detection frameworks that intercept ransomware during its early stages. For example, systems designed to monitor and analyze network traffic in real time can detect anomalies indicative of ransomware attempting to establish connections with its command-and-control servers [72].

One prominent example of this technology is The Microsoft Incident Response team conducted an in-depth analysis of a ransomware attack, detailing how signature-based detection methods were employed to identify known malicious patterns. This case study emphasizes the importance of maintaining up-to-date signature databases to effectively recognize and mitigate ransomware threats.

5.1.2. Heuristic-based detection

Heuristic-based detection offers a flexible and adaptive approach to identifying ransomware, particularly in IoT environments where threats evolve rapidly. Unlike signature-based methods, which rely on pre-defined identifiers of known malware, heuristic detection evaluates behaviors and patterns that deviate from normal operations to flag potential threats [73].

The core of heuristic-based detection lies in feature extraction and analysis [74]. Features such as abnormal system calls, irregular file operations, and suspicious network communications are monitored and analyzed.

The detection process integrates static and dynamic analysis. In static analysis, the program's code is examined for suspicious elements such as

obfuscated routines, malicious API calls, or embedded encryption libraries. Dynamic analysis, on the other hand, observes the program's real-time behavior in controlled environments like sandboxes.

Heuristic detection often employs rule-based systems to define suspicious behaviors. A rule might, for instance, flag any program attempting to disable antivirus software or rapidly modify large numbers of files. However, static rules alone are limited in addressing modern, sophisticated malware. To overcome this, heuristic methods frequently incorporate machine learning algorithms. These algorithms are trained on large datasets of benign and malicious activities, enabling the system to learn and classify behaviors with greater precision [75].

Environment emulation is another critical component of heuristic detection. This technique creates a simulated operational environment to observe how a program behaves when executed. Many ransomware variants attempt to evade detection by delaying their execution or altering their behavior when running in virtualized environments. Emulation ensures that such evasive tactics are captured and analyzed, providing an additional layer of security.

In [76], researchers utilized heuristic-based methods to detect ransomware. By analyzing behavioral patterns and employing machine learning algorithms, the study achieved a classification accuracy of 96 %, demonstrating the effectiveness of heuristic approaches in identifying ransomware activities.

5.1.3. Behavioral analysis-based detection

Behavioral analysis-based detection represents a dynamic and adaptive approach to identifying ransomware within IoT environments by closely monitoring device behaviors and network interactions [77]. Unlike static analysis, which focuses on inspecting code or signatures, behavioral analysis emphasizes real-time observation of activity patterns. This approach enables the detection of ransomware strains, including those previously unseen, by analyzing their actions rather than relying on pre-identified code features. Maze ransomware, which exfiltrates data, was flagged by behavioral analysis through unusual network traffic [34]. Through the establishment of a baseline representing normal device and network behavior, deviations that suggest malicious intent—such as abrupt spikes in file access, anomalous data encryption operations, or unusual network traffic—can be promptly flagged and investigated.

The strength of behavioral analysis lies in its ability to adapt to the evolving nature of ransomware. It leverages real-time data to identify patterns of abnormal activity, regardless of how the ransomware disguises its code [78]. For instance, ransomware often exhibits distinct operational traits, such as repetitive file access, unauthorized attempts to overwrite or encrypt files, or unexpected changes to system configurations, minimizing damage and enabling timely countermeasures.

Recent advancements highlight the sophistication and effectiveness of behavioral analysis in ransomware detection. For example, one study tackled the challenge of processing high-dimensional IoT data by employing particle swarm optimization and genetic programming techniques. These methods optimize feature selection and enhance the analysis of API call sequences, improving early detection rates by identifying subtle behavioral anomalies that might otherwise go unnoticed [79]. Another innovative contribution focused on identifying distinct patterns in API call sequences, such as the repeated encryption of files or anomalous memory access operations, as reliable indicators of ransomware activity. By using these patterns to differentiate between malicious and benign behaviors, the detection system can reduce false positives while maintaining high sensitivity [80].

Deep learning has further enriched the potential of behavioral analysis. A particularly notable advancement involves the use of deep neural networks to analyze API call sequences in a manner that not only enhances detection accuracy but also provides explainable insights into the decision-making process. By leveraging the interpretability of these models, cybersecurity professionals can better understand why certain behaviors are flagged as suspicious, facilitating more informed

responses to potential threats [81].

5.1.4. Artificial intelligence-based detection

Artificial intelligence (AI)-based detection techniques offer a dynamic and proactive approach to cybersecurity by leveraging advanced computational methods to predict and identify ransomware attacks. By analyzing large datasets and learning from past events, AI-based systems can detect subtle patterns or anomalies that may elude human analysts or traditional automated systems. These techniques are particularly advantageous in IoT environments, where scalability and adaptability are essential for managing diverse and rapidly evolving threats [82]. For instance, Ryuk ransomware was detected with AI-based methods like CNNs, which identified hidden patterns in encryption logs [32].

A prominent example of AI application in ransomware detection is the use of convolutional neural networks (CNNs), which are highly effective in extracting hierarchical features from complex data. CNNs are particularly suited for tasks involving sequential or spatial data, such as analyzing system logs, network traffic patterns, or encrypted data streams [83]. For instance, [84] demonstrated how CNNs could convert raw ransomware data into image-like structures, enabling the detection of patterns indicative of malicious activities. The CNN architecture leverages convolutional layers to identify localized features, such as repetitive encryption commands or unusual file interactions, followed by fully connected layers that classify the observed behavior as either malicious or benign. This hierarchical learning process allows CNNs to uncover intricate patterns that might be imperceptible to traditional detection methods.

The primary advantage of CNNs lies in their ability to automatically learn relevant features without requiring extensive manual feature engineering, making them particularly effective for ransomware detection in complex IoT environments [85]. Additionally, CNNs exhibit robustness to noise in input data, which enhances their accuracy when dealing with diverse ransomware samples. However, their performance is contingent upon the availability of large, labeled datasets for training, and they often require substantial computational resources, which can pose challenges in resource-constrained environments. Techniques such as transfer learning, where pre-trained models are fine-tuned on domain-specific data, and model pruning, which reduces the size of the network without significantly affecting performance, have been proposed to address these limitations [86].

Reinforcement learning (RL) is a promising AI technique in ransomware detection, capable of simulating attack-defense scenarios to learn optimal mitigation strategies. RL agents interact with their environment, receiving rewards or penalties to refine their responses. This adaptability enables RL-based systems to detect evolving threats like crypto-ransomware by analyzing real-time patterns in network traffic or device behavior, as demonstrated in [87].

A critical factor in RL's success is designing effective reward functions that guide the agent toward accurate and reliable detection without overfitting or bias. Poorly designed rewards can lead to sub-optimal strategies, such as over-detection of benign activities. Extensive training across diverse simulated scenarios is essential to ensure the agent generalizes well to unseen ransomware behaviors. While computationally intensive, RL's ability to dynamically adapt to new threats makes it a valuable tool in modern ransomware defense strategies.

In [88], researchers applied machine learning techniques to detect ransomware attacks. The study utilized the Hoeffding trees classifier, a stream data mining classification technique, to analyze and prevent ransomware activities, highlighting the potential of AI in enhancing cybersecurity measures.

5.1.5. Deception-Based detection

Deception-based techniques in ransomware detection embody a sophisticated strategy that focuses on diverting ransomware from legitimate targets using decoys and traps. These methods are rooted in creating highly realistic yet isolated environments, often referred to as

honeypots, which simulate actual network components, files, and folders. The purpose of these honeypots is to attract ransomware by imitating legitimate systems convincingly. By interacting with these decoys, ransomware inadvertently exposes its methods and triggers alerts, allowing cybersecurity teams to respond effectively [89].

One of the defining characteristics of deception-based techniques is their proactive nature. It prevents ransomware from accessing critical data while simultaneously providing valuable insights into the attack's structure. For instance, when ransomware interacts with decoy files or systems, it reveals its encryption methods, payload delivery mechanisms, and lateral movement patterns [90]. This information is critical for understanding emerging ransomware strategies and designing countermeasures. Security teams can analyze the captured data to identify attack vectors, assess the severity of the threat, and anticipate potential variations.

Recent innovations have expanded the scope and efficiency of deception-based techniques. For example, honey files—specialized decoy files embedded within a network—have proven effective in luring ransomware. When accessed, these files trigger alerts, enabling early detection and providing detailed behavioral data [91]. Systems like R-locker take this concept further by not only detecting interactions but also analyzing them in real time. R-locker uses these insights to mitigate ransomware activities, integrating detection and response into a streamlined process [92].

Another advancement involves the use of dynamic decoys. Unlike static decoy files or systems, dynamic decoys adapt to the evolving tactics of ransomware. They can alter their appearance, behavior, or network interactions to remain attractive and relevant to emerging ransomware variants. This adaptability enhances the effectiveness of deception-based techniques, ensuring that they remain a step ahead of attackers [93].

5.1.6. Sandboxing-based detection

Sandboxing-based detection is a highly effective cybersecurity approach that executes suspicious software within a secure, isolated virtual environment called a sandbox. This environment mimics real-world operating conditions, allowing potentially malicious programs, such as ransomware, to operate as they would on actual systems. By monitoring how the software behaves—such as file modifications, registry changes, or unusual network communications—the sandbox can identify malicious activities without exposing the main system to any risk [94].

Unlike static analysis, which focuses on examining the code of a program, sandboxing enables dynamic analysis by observing the real-time behavior of the software [95]. This makes it particularly effective against new and unknown ransomware variants that might bypass traditional detection methods. For example, ransomware often exhibits characteristic behaviors, such as rapidly encrypting files or communicating with command-and-control servers. A sandbox can detect these patterns early and trigger alerts before the ransomware can cause significant harm [96]. Modern sandboxes incorporate advanced measures, such as mimicking physical systems in a highly realistic manner or introducing randomness in execution to confuse the malware. These innovations help maintain the effectiveness of sandboxing, even against sophisticated threats.

Research and practical applications have demonstrated the significant role of sandboxing in modern cybersecurity. For instance, it has been successfully employed to analyze and detect advanced persistent threats (APTs), such as Flame and Red October, which use stealthy attack methods and complex payloads. Sandboxes also play a critical role in automated dynamic analysis, where large volumes of potentially malicious files are processed efficiently, enabling scalable ransomware detection [97].

The integration of sandboxing with machine learning and artificial intelligence has further enhanced its capabilities. These technologies enable sandboxes to learn from past threats and adapt to emerging ones,

improving detection accuracy over time. Additionally, combining sandboxing with broader security systems, such as endpoint protection and intrusion detection systems, creates a layered defense strategy that is more robust against ransomware.

5.1.7. Real-time protection systems-based detection

Real-time protection systems are critical for safeguarding IoT environments against ransomware by continuously monitoring system activities and responding to potential threats as they emerge [98]. Unlike traditional reactive approaches, these systems operate proactively, leveraging real-time data streams and sophisticated algorithms to detect and mitigate ransomware activities before they can inflict significant damage. By seamlessly integrating with network infrastructures, real-time protection systems enable continuous surveillance and rapid responses, making them particularly effective in dynamic and heterogeneous IoT environments.

Real-time protection systems primarily rely on anomaly detection and behavior analysis algorithms to identify malicious activities [99]. These systems monitor key indicators of ransomware behavior, such as rapid encryption requests, unusual file modifications, unauthorized network communications, or deviations from established usage patterns. By establishing a baseline of normal system behavior through historical data or pre-defined rules, the system can flag deviations as potential threats.

Advanced machine learning techniques, such as unsupervised clustering and deep learning, have been integrated into real-time protection systems to enhance detection accuracy [100]. For instance, unsupervised learning models like k-means clustering group data points based on similarity, identifying anomalies that fall outside established clusters. Similarly, recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have been employed to detect temporal and spatial patterns in system activities, allowing for the identification of ransomware behaviors that evolve over time.

Numerous studies have demonstrated the efficacy of real-time protection systems in detecting and mitigating ransomware attacks. For example, the ARDS system proposed in [101] employs anomaly detection algorithms to rapidly identify suspicious activities in IoT environments. This system not only detects deviations in real-time but also integrates a preemptive quarantine mechanism, isolating affected nodes to prevent the spread of ransomware. Similarly, Rwgward, introduced in [102], focuses on monitoring system-level behaviors indicative of encryption activities. By analyzing system calls and file access patterns, Rwgward can intercept ransomware in its early stages, effectively preventing file encryption before significant damage occurs.

5.1.8. File integrity monitoring-based detection

File Integrity Monitoring (FIM) is a cybersecurity technique designed to safeguard critical system files and configuration settings from unauthorized modifications, deletions, or encryptions, which are often early indicators of ransomware attacks [103]. The central principle of FIM involves maintaining a secure baseline of file integrity, continuously monitoring for deviations from this baseline, and promptly alerting administrators to potential threats. This proactive approach not only facilitates the early detection of ransomware but also prevents its execution by identifying suspicious activities before they can cause significant damage.

The core functionality of FIM lies in its ability to compare the current state of files with a known secure baseline. This baseline includes cryptographic checksums, access control lists, and metadata for critical files. Whenever a monitored file is altered, the FIM system generates an alert, allowing administrators to investigate whether the change was legitimate or malicious. FIM systems often employ hash algorithms like MD5, SHA-1, or SHA-256 to calculate checksums, ensuring high accuracy in detecting even minor modifications. In [104], researchers demonstrated the efficacy of FIM in identifying unauthorized changes to critical system files, emphasizing its role in issuing timely alerts to

contain threats before ransomware can propagate.

Deep learning methods have further advanced FIM capabilities by providing a more nuanced analysis of file activity patterns. In [105], researchers employed Bidirectional Long Short-Term Memory (BiLSTM) networks combined with Conditional Random Fields (CRFs) to analyze sequences of file access and modification events. BiLSTM models are particularly effective at capturing the temporal dependencies in sequential data, making them well-suited for identifying ransomware behaviors that unfold over time, such as the gradual encryption of files or the deletion of backup data. The addition of CRFs enables the system to make structured predictions, enhancing its ability to detect malicious activity with high precision. This approach demonstrated real-time detection of ransomware, preventing its execution and limiting the impact on critical systems.

5.1.9. Cloud-based detection

Cloud-based ransomware detection leverages the scalability and flexibility of cloud computing to provide robust defenses against ransomware threats in distributed IoT networks [106]. By centralizing the analysis of data collected from diverse devices, cloud-based systems enable comprehensive and rapid threat detection, ensuring timely responses to mitigate attacks. These systems utilize advanced analytics, threat intelligence, and machine learning models deployed on the cloud to identify ransomware patterns and anomalies in real time. Additionally, automated backup and recovery mechanisms are often incorporated to safeguard critical data, allowing for seamless restoration without succumbing to ransom demands [107].

One of the primary strengths of cloud-based ransomware detection lies in its ability to process large volumes of heterogeneous data. The centralized nature of cloud platforms allows for the aggregation of logs, telemetry, and system behaviors from numerous IoT devices, creating a unified dataset for analysis. Advanced machine learning models deployed on the cloud, such as convolutional neural networks (CNNs) and transformers, can extract complex features and detect ransomware activities that traditional rule-based methods might overlook. For instance, the RANSOMNET+ model proposed in [108] integrates CNNs and transformers to analyze encrypted cloud data, enabling the detection of ransomware by identifying subtle patterns in the data's structure. This approach combines the localized feature extraction capabilities of CNNs with the global contextual understanding of transformers, making it particularly effective in scenarios involving encrypted or high-dimensional data.

Another key innovation in this field involves analyzing volatile memory features in private clouds, as discussed in [109]. This approach leverages machine learning algorithms to detect ransomware by examining temporary memory states, which often reveal encryption activities or malicious payload executions. By focusing on memory patterns, this method enhances detection accuracy and reliability, even in cases where ransomware employs obfuscation techniques to evade static analysis. Furthermore, it reduces dependency on signature-based methods, making it more resilient to emerging and zero-day threats.

CloudRPS, described in [110], exemplifies the integration of real-time data analysis with automated backup solutions. By monitoring data interactions and identifying anomalies indicative of ransomware attacks, this system preemptively isolates affected components while maintaining seamless user access to cloud backups. This dual functionality not only minimizes the impact of ransomware but also reduces the computational load on individual devices, shifting resource-intensive tasks to the cloud.

The authors of [111] explored the implementation of cloud-based detection systems. By leveraging the scalability and computational power of cloud platforms, the research highlights how machine learning models can be deployed to analyze large datasets for ransomware detection, providing timely and efficient threat identification.

5.1.10. Blockchain-based detection

Blockchain-based detection methods are an innovative approach to combating ransomware by utilizing the core strengths of blockchain technology: immutability, decentralization, and transparency [112]. These attributes create a highly secure environment that can detect and prevent ransomware attacks by ensuring data integrity and providing a verifiable, tamper-resistant audit trail. It is proved effective against distributed ransomware like Cerber by using immutable hash comparisons to detect unauthorized encryption [29].

The immutability of blockchain ensures that once data is recorded, it cannot be altered or deleted. This characteristic is particularly valuable in ransomware detection [113]. Any unauthorized modification or encryption attempt becomes evident, as the original data and its cryptographic hash stored on the blockchain cannot be tampered with. For example, blockchain-based systems can compare the hash of a file against its recorded hash on the blockchain. If the hashes do not match, it signals potential ransomware activity.

Blockchain’s decentralized nature further enhances its effectiveness in ransomware defense [114]. In a blockchain network, data is distributed across multiple nodes, and each node holds a complete copy of the blockchain. This distribution eliminates single points of failure, a common vulnerability in traditional systems. If ransomware compromises a single node, the remaining nodes maintain intact copies of the data, ensuring the continuity of operations and data integrity. Decentralization also makes it significantly harder for ransomware to encrypt or manipulate data without being detected across the network.

In IoT ecosystems, blockchain can play a critical role in ensuring secure and reliable data handling. IoT devices are often targeted by ransomware due to their limited built-in security measures and centralized control. Blockchain can enhance their security by providing a decentralized framework for data validation. For instance, every interaction or data transaction within an IoT network can be logged on the blockchain, creating a transparent and immutable record. If a ransomware attack attempts to encrypt or alter IoT data, the blockchain’s consistency checks across nodes can detect the anomaly and prevent the attack from spreading [115].

5.2. Discussion

Table 4 summarizes the advantages and disadvantages of each ransomware detection technique based on the above content. Signature-based detection is highly effective for recognizing known threats by utilizing predefined patterns or signatures, making it straightforward and resource-efficient. However, its effectiveness is limited against new or unknown threats unless its signature database is regularly updated [. Heuristic-based detection evaluates the behavior of files and programs to identify suspicious activities, enabling it to detect malware that signature-based methods might miss. While this approach excels in identifying new threats, it is prone to false positives, potentially flagging legitimate software as malicious. This limitation demands more sophisticated algorithms and increases computational resource requirements. Behavioral analysis-based detection continuously monitors system behaviors to identify anomalies indicative of a ransomware attack, such as unusual encryption activities. Although this method is powerful, it requires a well-defined baseline of normal behavior, which can make it resource-intensive and susceptible to false alarms.

Artificial Intelligence (AI)-based detection employs advanced algorithms, such as machine learning, to process vast amounts of data and identify subtle patterns that may elude human analysts. These systems improve as they process more data but demand significant computational resources and large datasets for effective training, posing a substantial initial investment. Deception-based detection deploys decoys to trap ransomware by simulating vulnerable targets, effectively diverting threats away from critical assets. Although this approach provides valuable insights into attacker behavior and strategies, it is complex to implement and manage. Sandboxing involves executing suspicious

Table 4
Overview of advantages and disadvantages of ransomware detection techniques on IOT.

Techniques	Advantages	Disadvantages
Signature-based Detection	Effective at recognizing known threats using predefined patterns Straightforward and resource-efficient	Limited effectiveness against new, unknown threats unless updated regularly
Heuristic-based Detection	Can catch malware that signature methods might miss by evaluating behavior	May mistakenly flag legitimate software as malicious Demands more sophisticated programming
Behavioral Analysis-based Detection	Monitors system behaviors to spot anomalies, powerful detection capability	Requires a well-defined baseline of normal behavior Resource-intensive and prone to false alarms
Artificial Intelligence-based Detection	Employs machine learning to identify subtle patterns from large data sets Improves over time	Requires significant computational power and data for effective training Substantial initial investment
Deception-based Detection	Uses decoys to trap ransomware, providing insights into attack methods Diverts threats from real assets	Complex to implement and manage
Sandboxing-based detection	Runs suspicious programs in isolated environments without risking system safety	Can increase processing overhead and slow down system performance
Real-Time Protection Systems-based detection	Constantly monitors and responds to threats, highly effective	Demands significant resources for continuous operation
File Integrity Monitoring-based detection	Focuses on protecting against unauthorized changes, crucial for early detection	Might struggle against sophisticated attacks that bypass standard monitoring
Cloud-Based detection	Utilizes cloud computing for scalable and flexible threat management across networks	Can introduce latency Depends heavily on continuous internet access
Blockchain-Based detection	Leverages security features like decentralization and tamper-resistant ledgers	Still a novel application in cybersecurity with no mature standards

programs in a controlled, isolated environment to observe their behavior without risking the safety of actual systems. While this proactive layer of security is effective, it increases processing overhead and may slow down overall system performance. Real-time protection systems monitor and respond to threats as they occur, providing immediate defense against attacks. Despite their high effectiveness, these systems require significant computational resources for continuous operation.

File Integrity Monitoring (FIM) focuses on detecting unauthorized changes to files, which is crucial for identifying ransomware attacks at an early stage. FIM can quickly detect deviations from a secure baseline; however, it may struggle against advanced attacks designed to bypass standard monitoring techniques. Cloud-based detection leverages the scalability and flexibility of cloud computing to manage threats effectively across distributed networks. This approach offers robust threat management but may introduce latency and relies heavily on uninterrupted internet connectivity. Blockchain-based detection utilizes the inherent security features of blockchain technology, such as decentralization and tamper-resistant ledgers, to enhance data security. While promising, the application of blockchain in cybersecurity remains relatively novel and lacks mature standards and practices. Each method presents unique strengths and challenges, making the choice dependent on specific security needs, environmental complexity, and available resources.

To compare various ransomware detection methods effectively, it is essential to establish appropriate evaluation criteria. For this purpose, we conducted a comprehensive analysis of these detection technologies across five key dimensions: accuracy, speed, rapid response, scalability,

and robustness, as summarized in Table 5. Accuracy refers to the ability of a detection method to correctly identify ransomware, minimizing both false positives and false negatives. Speed measures the time required for detection and response, which is critical for mitigating the potential impact of ransomware attacks. Rapid response evaluates the method's ability to act swiftly to contain damage. Scalability assesses the method's performance in handling large volumes of data or operating in extensive networks, particularly significant in IoT environments. Robustness refers to the resilience of the detection method against evasion techniques and its effectiveness in identifying sophisticated or novel ransomware variants. Additionally, usability considers the ease of implementation and management, including user experience and maintenance requirements. These criteria were selected to provide a balanced evaluation of detection methods, considering both technical performance and practical feasibility.

Based on the classification of ransomware detection methods, a detailed comparison among all categories can be made. As shown in Table 5, signature-based detection is highly accurate for known threats but struggles against novel ransomware due to its reliance on predefined signatures. Heuristic-based, behavioral analysis-based, and AI-based detection methods offer varying levels of accuracy by analyzing patterns, real-time interactions, and advanced algorithms. Heuristic-based methods identify anomalies through pattern recognition, striking a balance between speed and precision. Behavioral analysis-based approaches monitor real-time activities to detect deviations from normal behavior, demonstrating high accuracy against sophisticated threats. AI-based systems leverage machine learning algorithms that evolve over time, making them particularly effective against new ransomware variants.

Deception-based, sandboxing, and real-time protection systems employ unique monitoring techniques to detect threats. Deception-based methods create decoys to lure ransomware, providing valuable insights into attack strategies. Sandboxing isolates potentially malicious programs in controlled environments, enabling behavioral analysis without endangering the main system. Real-time protection systems continuously monitor ongoing activities, offering instant responses to detected threats to ensure system security. File integrity monitoring safeguards systems by verifying file integrity and detecting unauthorized changes at an early stage. Cloud-based detection methods utilize the scalability of cloud computing to analyze large datasets efficiently and manage threats across distributed networks. Blockchain-based methods enhance security by using immutable ledgers for secure data logging, preventing tampering and ensuring data integrity.

However, each detection method has limitations. Signature-based detection, while fast and resource-efficient, depends heavily on predefined data and struggles with new threats, requiring frequent updates to remain effective. Heuristic-based and behavioral analysis-based methods demand substantial real-time processing resources, posing challenges to scalability. AI-based detection requires significant data infrastructure and expertise, making its deployment resource-intensive. Deception-based and sandboxing methods involve complex implementation and ongoing maintenance. Real-time protection, cloud-based,

and blockchain-based systems require intricate setups and specialized knowledge to achieve optimal performance. Each detection method presents specific implementation challenges, highlighting the need for continuous improvement to address the evolving nature of ransomware threats.

In summary, each ransomware detection method demonstrates unique strengths and weaknesses across the five evaluation criteria: accuracy, speed, scalability, robustness, and usability. Signature-based detection excels in speed and usability due to its straightforward implementation but lacks robustness against novel threats. Heuristic-based detection strikes a balance between speed and scalability while offering moderate accuracy. Behavioral analysis-based detection achieves high levels of accuracy and robustness but encounters challenges in scalability and usability due to its resource-intensive nature. AI-based detection delivers excellent accuracy and robustness with strong scalability, although its complexity makes implementation demanding.

Deception-based detection stands out for its robustness and quick response, but it is challenging to manage and maintain. Sandboxing-based detection effectively combines accuracy and robustness but offers only moderate speed. Real-time protection systems ensure high accuracy and robustness with strong scalability, providing effective defenses in dynamic environments. File integrity monitoring-based detection is accurate and robust, yet it faces significant challenges in scalability and management.

Cloud-based detection excels in scalability, accuracy, and robustness by leveraging the flexibility of cloud computing but introduces complexities in implementation and maintenance. Blockchain-based detection offers high accuracy, robustness, and scalability, utilizing its inherent security features, though its deployment remains complex and resource-intensive. These observations highlight the trade-offs inherent in each detection method, emphasizing the importance of selecting approaches based on specific security needs and resource constraints.

This chapter presented multiple detection methods, each with unique strengths and limitations across different IoT scenarios. Before we consider how to practically counter these threats, we must further explore the range of countermeasures available. The next chapter will discuss prevention and protection strategies that can mitigate ransomware's impact.

6. Countermeasures

Countermeasures refer to strategies, tools, or practices designed to mitigate cybersecurity threats, including ransomware, as illustrated in Fig. 7. These measures enhance security by preventing attacks, reducing the impact of vulnerabilities, and facilitating rapid recovery. In the context of ransomware, countermeasures aim to block ransomware from infiltrating systems, detect anomalous activities indicative of an attack, contain the spread of the infection by isolating compromised systems, and ensure that secure backups enable recovery without succumbing to ransom demands. Implementing robust ransomware countermeasures allows organizations to preserve data integrity and minimize operational disruptions. The following section presents a classification of

Table 5
Compassion of ransomware detection techniques in terms of [152–172].

Techniques	Accuracy	Speed	Scalability	Robustness	Usability
Signature-based detection	High	High	Moderate	Moderate	High
Heuristic-based detection	Moderate	Moderate	High	Moderate	Moderate
Behavioral analysis-based detection	High	Moderate	Moderate	High	Moderate
AI-based detection	High	High	High	High	Moderate
Deception-based detection	Moderate	High	Moderate	High	Moderate
Sandboxing-based detection	High	Moderate	High	High	Moderate
Real-time protection systems	High	High	High	High	Moderate
File integrity monitoring-based detection	High	Moderate	Moderate	High	Moderate
Cloud-based detection	High	High	High	High	Moderate
Blockchain-based detection	High	Moderate	High	High	Moderate

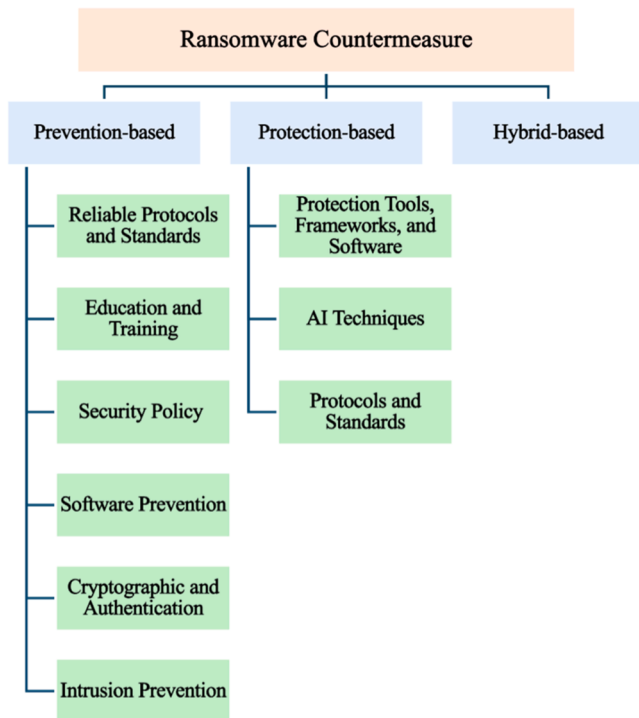


Fig. 7. Classification of ransomware attacks countermeasure.

countermeasures specifically targeting ransomware attacks.

6.1. Classification

6.1.1. Prevention-based

Prevention-based countermeasures are designed to thwart ransomware before it gains access to systems [116]. They encompass a broad range of proactive strategies, including the deployment of advanced antivirus software, the application of reliable protocols and standards to secure data exchange, and the continuous education of employees on the latest cyber threats. These countermeasures are frequently organized into six main categories—Reliable Protocols and Standards, Education and Training, Security Policy, Software Prevention, Cryptographic and Authentication, and Intrusion Prevention—each of which contributes to a structured defensive posture. Although these categories sometimes overlap, they collectively stress the importance of anticipating potential threats, elevating security awareness, and ensuring that systems are resilient to new forms of ransomware. Yet, they face ongoing challenges in consistent enforcement and in keeping pace with rapidly evolving attack methods.

6.1.1.1. Reliable protocols and standards. Reliable protocols and standards establish recognized guidelines and technical specifications that help organizations maintain secure data exchanges, both internally and externally [117]. Industry-standard protocols, such as SSL, TLS, and IPSec, have been repeatedly proven to protect data in transit through robust encryption. Studies examining the evolution and implementation of these protocols outline their effectiveness in electronic communications and e-commerce, while also highlighting potential vulnerabilities [118]. Ongoing compliance checks, periodic protocol updates, and targeted training for employees on how to implement and follow these standards ensure that security measures remain both relevant and effective. However, organizations must continuously evaluate how new threat vectors—such as more advanced ransomware variants—can exploit misconfigurations or outdated protocol versions, a risk that underscores the importance of regular reviews and a dynamic upgrade process.

6.1.1.2. Education and training. Education and training concentrate on building a cybersecurity-aware culture in which employees understand how to recognize suspicious links, handle sensitive data, and report unusual activity [119]. Several studies confirm the efficacy of regular training sessions and workshops, which update personnel on the latest phishing techniques and other social engineering tactics frequently used in ransomware attacks [120]. Tailored programs that account for different roles, levels of technical knowledge, or even demographic factors such as age or gender can improve employees' response to emerging threats. Researchers also note the need to move beyond mere compliance-based instruction; integrating social marketing techniques and practical exercises—like simulated phishing campaigns—can create more enduring behavioral change. Despite these efforts, organizations can still encounter low employee engagement, limited budget, or insufficient managerial support, all of which undermine the potential of education and training to serve as a robust preventive layer.

6.1.1.3. Security policy. A well-defined security policy offers a formal framework for safeguarding an organization's information technology assets and for ensuring that all departments maintain a uniform security posture [121]. Such a policy generally covers rules on encryption, guidelines for handling suspicious emails, and procedures for incident reporting. Previous research provides both conceptual frameworks and practical case analyses that detail how organizations can design, implement, and enforce effective security policies to manage threats [122]. However, even the most comprehensive policies demand continuous updates to remain aligned with emerging attack vectors, and they require consistent enforcement across all levels of the organization. Without genuine leadership commitment or clear communication about the policy's rationale, employees may not fully comply with or understand these guidelines, leaving gaps for ransomware intrusions.

6.1.1.4. Software prevention. Software prevention focuses on using anti-ransomware, antivirus software, and other detection tools to identify, block, and remove potentially malicious programs, including ransomware [123]. Reliability and proper configuration are vital, since even a highly rated security product may fail if it is poorly tuned or not updated regularly. Over time, attackers have developed sophisticated techniques, such as fileless ransomware or advanced persistence mechanisms, that can bypass traditional antivirus detection. Consequently, researchers have explored various machine learning approaches to enhance ransomware detection, although such systems may generate false positives and often require continuous re-training to keep pace with emerging threats [124]. Industry guidance, including best practices from Microsoft, emphasizes the importance of mapping critical applications, maintaining system integrity, and isolating infected machines to limit the spread of ransomware. Despite these advances, software-based prevention cannot operate as a standalone solution. It must be integrated with broader cybersecurity policies, training, and alert mechanisms to minimize blind spots.

6.1.1.5. Cryptographic and authentication. Strong cryptographic algorithms and authentication mechanisms help protect sensitive data by limiting access to authorized users [125]. Organizations often rely on techniques such as AES and RSA for data encryption at rest and in transit, while multi-factor authentication (MFA) adds another layer of security for system logins. Research has explored hybrid encryption methods that combine algorithms like Advanced Encryption Standard (AES) with Elliptic Curve Cryptography (ECC) to enhance the security and efficiency of IoT networks [126]. Although these cryptographic approaches can significantly reduce the risk of data exposure or unauthorized modifications, they also introduce challenges related to key management, computational overhead, and the need to stay updated with evolving cryptographic standards that address newly discovered vulnerabilities. Regular reviews of cryptographic protocols and key

rotation practices are therefore essential to maintain resilience.

6.1.1.6. Intrusion prevention. Intrusion Prevention Systems (IPS) monitor network and system activities in real-time to detect and block malicious behavior before it can compromise other segments of the infrastructure [127]. Proper deployment involves choosing an IPS suited to organizational needs and configuring it with current threat signatures, as well as behavioral detection rules capable of identifying anomalies associated with new ransomware strains. Recent studies have proposed machine learning-driven improvements in IPS, and adaptive deep learning approaches like LeNet-5 demonstrate increased accuracy in detecting abnormal traffic patterns [128]. While these advances offer promising detection rates, they require frequent signature updates and training data that reflect the shifting cyber-threat landscape. Furthermore, intrusion prevention may generate false alarms that overwhelm security teams, thus emphasizing the need for complementary layers of defense and a risk-based approach to IPS deployment.

6.1.2. Protection-based

When preventive measures are not fully effective or fail to address newly emerging threats, protection-based strategies serve as a secondary barrier to mitigate harm [129]. Typically, these defenses encompass Intrusion Detection Systems (IDS) for early-warning monitoring, robust data backups to allow swift system restoration, and network segmentation to limit lateral movement. While these measures are often considered passive, they can provide timely containment of ransomware attacks, reducing the likelihood of catastrophic data loss. Nonetheless, the success of protection-based countermeasures can be undermined if they are not rigorously tested or if backups are irregularly performed and stored in insecure locations.

6.1.2.1. Protection tools, frameworks, and software. A variety of specialized tools, frameworks, and software solutions focus on containing the damage once an attack is underway, and they typically include antivirus programs, firewalls, and timely security updates [130]. Studies of ransomware on Windows systems underscore the importance of both signature-based detection and more proactive methods such as behavior analysis and machine learning [131]. Firewalls serve as a gatekeeper for network traffic, but they must be combined with efficient patch management, as attackers frequently exploit unpatched vulnerabilities to distribute ransomware [132]. These strategies are most effective if they are integrated into a broader incident response plan, so that organizations can quickly identify the scope of an infection, isolate compromised segments, and restore systems from clean backups. However, heavy reliance on automated solutions can sometimes create a false sense of security if users do not consistently update signatures or if the organization lacks a well-structured incident response procedure.

6.1.2.2. AI techniques. Artificial intelligence (AI) stands out for its ability to analyze large volumes of data and identify subtle anomalies that traditional signature-based methods might miss. Machine learning and deep learning solutions can detect emerging attack patterns by examining historical records of malicious behaviors, while reinforcement learning allows adaptive responses based on the consequences of previous actions [133]. In IoT environments, ML-based systems have been shown to spot malicious activities in near-real-time, and deep learning can improve detection rates for ransomware encrypted at different layers of the network. Reinforcement learning additionally offers self-adjusting security policies, making it harder for attackers to predict and bypass static defenses [134]. Game theory provides yet another perspective by modeling the interplay between attackers and defenders, shedding light on cost-effective strategies that maximize deterrence. Yet, these AI-driven defenses often rely on substantial datasets that must be continuously updated, and they may overlook highly sophisticated adversarial techniques designed to evade pattern

recognition. Integrating AI within a layered security framework is thus essential for maintaining a more holistic defense.

6.1.2.3. Protocols and standards. In addition to conventional protective tools, protocols and standards specifically tailored for post-infection scenarios can strengthen overall resilience to ransomware. Enforcing secure communication protocols (e.g., SSL or IKEv2) and robust authentication procedures becomes vital once a network intrusion is detected. Past experiences with protocol design, including the need to learn from cryptographic weaknesses or configuration errors, show how good protocol hygiene can minimize exploitable vulnerabilities [135, 136]. These measures alone, however, offer limited mitigation if the attacker has already penetrated deeply. Consistent protocol updates, real-time threat intelligence, and restricted access privileges for sensitive data are necessary complements, ensuring that attackers can neither escalate privileges nor move freely through a network after an initial breach.

6.1.3. Hybrid-based

Hybrid-based approaches merge preventive and protective techniques, aiming to create a multi-layered defense that can anticipate, detect, and respond to threats in real time. By combining predictive analytics and automated incident response, these approaches can stop nascent ransomware attempts and minimize disruptions if an attack slips through preventive measures. Recent work illustrates the limitations of existing cybersecurity tactics and proposes layered solutions that incorporate phishing protection, automated response systems, and cyber insurance to offset financial risks [137]. Other studies integrate static and dynamic analyses of cyberattacks with machine learning to improve detection speed and accuracy, particularly in large enterprises or critical infrastructures [138]. This multi-pronged approach also includes mechanisms to detect and block malicious emails, defend against cloud-based threats, and implement zero-trust principles to restrict lateral movement [139]. Although these comprehensive frameworks are more resource-intensive and demand highly skilled security teams, they offer a viable strategy against increasingly adaptive ransomware variants. By uniting different defensive elements—from employee training and careful protocol enforcement to AI-driven anomaly detection—organizations can cultivate a deeper resilience against both current and future ransomware threats.

6.2. Discussion

Based on our previous discussion, the advantages and disadvantages of each countermeasure approach are summarized in Table 6. Reliable protocols and standards provide a foundational layer of security by ensuring regular updates and compliance checks to address evolving threats and technologies. However, this approach demands continuous monitoring and frequent updates, which can be resource-intensive and may fail to address novel or sophisticated cyber threats effectively. Education and training mitigate the risk of human error by improving employees' cybersecurity awareness and practices. Nevertheless, maintaining relevance and engagement in training programs requires ongoing effort and resources. Moreover, the effectiveness of these programs largely depends on employee participation and their ability to retain the knowledge imparted. Security policies establish clear protocols and guidelines for managing and protecting IT resources, thereby strengthening the organization's baseline security posture. However, such policies must be regularly updated and consistently enforced across all organizational levels to remain effective.

Software-based prevention measures, such as anti-ransomware and antivirus solutions, proactively detect and block threats. These tools require constant updates and proper configuration to ensure efficacy, which may result in false positives or failure to detect new, unknown ransomware variants. Encryption and secure access controls further

Table 6
Overview of advantages and disadvantages of countermeasure.

Techniques	Advantages	Disadvantages
Reliable Protocols and Standards	Security foundation via protocols like SSL, TLS Adapts to evolving threats with regular updates	Resource-intensive with continuous monitoring May not effectively counter complex threats alone
Education and Training	Reduces human error by enhancing cybersecurity knowledge Tailored to specific organizational roles	Requires ongoing resource commitment Effectiveness depends on employee engagement
Security Policy	Provides clear IT asset management guidelines Facilitates a structured response to incidents	Needs regular updates and organizational support May be viewed as restrictive by employees
Software Prevention	Blocks threats with anti-ransomware and antivirus solutions Adapts to new threats with updates	Requires consistent updates and correct configuration Potential for false positives
Encryption and Authentication	Ensures data confidentiality and integrity Restricts access to authorized users	Complex implementation and management Requires careful key and security practices
Intrusion Prevention	Enhances security by automatically detecting and blocking threats Tailorable to specific network needs	Needs regular updates to maintain effectiveness Significant resource requirements for maintenance
Protection Tools, Frameworks, and Software	Reduces attack impact with systems like intrusion detection Includes data backups for recovery	Primarily reactive, not preventive Dependent on the effectiveness of other security measures
AI Techniques	Proactive with intelligent threat detection Adapts to new threats dynamically	High technical and resource requirements Potential data privacy and processing issues
Protocols and Standards	Enhances security through data encryption Restricts access to data for unauthorized users	Requires extensive technical expertise and resources Challenging to ensure consistency across large orgs
Hybrid-based Measures	Provides comprehensive, multi-layered defense Flexible response to various threats	Complex implementation and management Significant financial investment required

enhance data confidentiality and integrity. However, implementing and managing encryption protocols is complex, requiring robust key management and stringent security practices. The automatic detection and blocking of potential threats improve overall network security by leveraging predefined signatures and rules. This method, however, necessitates frequent updates to remain effective against emerging threats and demands significant resources for proper configuration and maintenance.

Protection tools, frameworks, and software aim to limit damage when preventive measures fail. These include intrusion detection systems, data backups, and network segmentation. Protocols and standards enhance network security by encrypting data and restricting access to authorized users. Nonetheless, implementing these measures requires considerable technical expertise and resources, and achieving consistency across large organizations remains a significant challenge due to protocol variations. AI-based technologies employ machine learning and deep learning to enhance ransomware detection and response capabilities through advanced analytics and real-time monitoring. These intelligent systems proactively detect and respond to threats but demand substantial expertise and infrastructure investment. Finally, a hybrid approach integrates multiple security methodologies, offering flexibility and a balanced combination of proactive and reactive strategies. Despite

its advantages, this approach involves the complexity of integrating diverse technologies and tactics, alongside considerable financial and managerial investment.

When evaluating ransomware countermeasures, various strategies demonstrate distinct strengths and limitations across key metrics such as accuracy, speed, scalability, robustness, and usability, as summarized in [Table 7](#). Reliable protocols and standards, such as SSL (Secure Sockets Layer) and TLS (Transport Layer Security), are highly effective in ensuring accurate and secure data transmission. However, their implementation and maintenance demand significant technical expertise. Education and training programs improve cybersecurity awareness and reduce the likelihood of human error. These programs, while effective, depend heavily on sustained employee engagement and the continuous updating of training materials to remain relevant.

Security policies enable the consistent application of security measures across an organization, providing high scalability and robustness. Despite their advantages, their usability may be moderate, as policies must be meticulously designed and regularly updated to address emerging threats. Software-based prevention methods, such as anti-ransomware and antivirus solutions, excel in accuracy and real-time threat detection. However, their efficacy against novel ransomware threats depends on the timely and frequent updating of their databases. Cryptographic and authentication measures enhance the security of data access and transmission by offering high levels of robustness and scalability. These measures, however, require careful key management and the proper implementation of authentication protocols to maintain their effectiveness.

Intrusion prevention systems (IPS) accurately detect and block known threats in real-time, further bolstering robustness, especially when integrated with machine learning algorithms. Protection tools, frameworks, and software, such as firewalls and antivirus programs, are generally user-friendly and scalable. Yet, their performance necessitates ongoing maintenance to ensure effectiveness. AI-based techniques bring high accuracy and adaptability, enabling real-time detection and proactive responses to threats. Nonetheless, these systems require considerable technical expertise and infrastructure for successful deployment and operation. Protocols and standards, while essential for enhancing network security, primarily function as preventive measures and do not directly detect or respond to threats. Finally, hybrid-based approaches combine multiple countermeasures to provide comprehensive protection. These approaches are characterized by high accuracy, scalability, and robustness, but their implementation is often complex due to the need to integrate diverse technologies and methodologies effectively.

We explored a variety of ransomware countermeasures, illustrating how they can be adapted to healthcare, manufacturing, and urban infrastructure. Although these strategies form a strong defense, challenges remain as attackers continually adapt. The next chapter will address these lingering issues and offer insights into future directions for research and collaboration.

7. Challenges and future directions

Looking at the trajectory of ransomware attacks and defenses, both attackers and defenders are poised to adopt innovative technologies and methods in the coming years. Attackers aim to evade increasingly sophisticated security measures, while defenders strive to detect and mitigate threats with minimal disruption to business operations. The interplay between offensive and defensive innovation will shape the future of ransomware threats.

On the offensive side, one notable trend is the integration of artificial intelligence (AI) and machine learning (ML) techniques into ransomware design [140]. Just as defenders leverage AI for anomaly detection, attackers may use AI to craft highly personalized phishing emails or to dynamically alter ransomware signatures, making conventional signature-based detection less effective. Another possibility is the rise of AI-driven reconnaissance, where attackers automate the collection of

Table 7
Compassion of ransomware countermeasures in Terms of [218–225].

Techniques	Accuracy	Speed	Scalability	Robustness	Usability
Reliable Protocols and Standards	High	Moderate	High	Moderate	Moderate
Education and Training	Moderate	Low	Moderate	Moderate	High
Security Policy	High	Moderate	High	High	Moderate
Software Prevention	High	High	High	Moderate	
Cryptographic and Authentication	High	Moderate	High	High	Moderate
Intrusion Prevention	High	High	High	Moderate	Moderate
Protection Tools, Frameworks, and Software	High	Moderate	High	Moderate	High
AI Techniques	High	High	High	High	Moderate
Protocols and Standards	High	Moderate	High	High	Moderate
Hybrid-based	High	High	High	High	Moderate

information about potential targets, including their software stack, user behavior patterns, and network configurations. With this intelligence, they can tailor ransomware payloads more accurately and launch targeted campaigns that exploit zero-day vulnerabilities or niche configurations.

Moreover, the continued growth of fileless ransomware and ‘living off the land’ strategies will likely reduce the footprint of ransomware within a target system [141]. By using legitimate system utilities and memory-resident techniques, adversaries can circumvent many traditional endpoint security products, especially those relying on static scans or sandboxing. Attackers may also target supply chains—particularly software updates and cloud services—to infiltrate businesses and deliver ransomware more efficiently. In parallel, developments in stealth encryption algorithms or advanced packing tools could further hinder static and behavior-based detection methods, forcing security tools to rely on more intensive data analysis and real-time monitoring of system events.

A more distant, yet significant, consideration is the potential impact of quantum computing [142]. Although large-scale quantum decryption capabilities remain speculative today, advances in quantum algorithms could eventually threaten widely used public-key cryptosystems like RSA or ECC. Forward-looking attackers may, in the future, invest in post-quantum cryptography research to break or weaken existing encryption methods, or they may hold encrypted data until quantum decryption becomes feasible, leading to new extortion models based on prolonged data exposure.

On the defensive side, organizations are increasingly deploying advanced AI and ML systems to detect anomalies that slip past static defenses. These systems rely on large-scale data collection across endpoints, network flows, and user behavior, often leveraging deep learning to identify subtle changes in system processes indicative of malicious encryption or lateral movement. Machine learning models are also adapting to novel ransomware strains by using behavioral data rather than fixed signatures, which helps identify suspicious events even when file hashes or signatures are unknown.

In parallel with AI-driven detection, defenders are turning to zero-trust architectures to minimize the risks posed by successful intrusions [143]. Zero-trust principles, which often rely on micro-segmentation and continuous authentication, limit how far ransomware can spread inside a compromised network. Cloud service providers and large enterprises are incorporating hardware-based security features, such as trusted platform modules or secure enclaves, to safeguard critical data and keys from malicious processes. Beyond that, blockchain-based solutions have been proposed for more tamper-proof logging and auditing, though their adoption for large-scale enterprise security is still evolving [144].

Finally, defenders are placing greater emphasis on automated incident response [145]. As AI mechanisms gain confidence, they can execute isolation or remediation steps in real time—suspending suspicious processes, quarantining infected systems, or reverting to trusted backups automatically. This automation aims to curtail damage within seconds or minutes of detecting an anomaly, thus reducing the window

during which ransomware can encrypt files. However, implementing such a system demands careful calibration of false positives to avoid interrupting normal business operations.

In summary, attackers will likely exploit AI for more personalized, stealthy, and automated ransomware campaigns, potentially integrate quantum-resistant or quantum-exploiting cryptography in the long term and continue refining fileless and supply-chain infiltration techniques. Defenders will respond with increasingly sophisticated AI-driven analytics, zero-trust network segmentation, hardware-backed security modules, and automated incident response to impede ransomware operations at every phase of an attack. The outcome of this escalating technological arms race will depend not only on advances in computational and cryptographic methods but also on organizations’ willingness to invest in continuous monitoring, rapid patching, and security-by-design principles.

In this chapter, we discussed the ongoing evolution of ransomware and the core challenges facing IoT security, considering potential solutions like blockchain and AI. However, effectively countering these threats requires broader, interdisciplinary coordination. In the next and final chapter, we summarize our findings and present key recommendations to guide both research and practical implementation.

8. Conclusions

This study provides a comprehensive analysis of the multifaceted characteristics of ransomware attacks on Internet of Things (IoT) devices, highlighting the pressing need for robust detection and countermeasure strategies. Ransomware has evolved significantly, ranging from simple locking mechanisms to sophisticated cryptographic and hybrid attacks, reflecting the increasingly severe challenges confronting cybersecurity professionals. The research underscores the pivotal role of diverse detection technologies—spanning from traditional signature-based methods to advanced artificial intelligence (AI)-driven approaches—in identifying and mitigating ransomware threats. While each technology offers distinct advantages, they also exhibit limitations, particularly when addressing novel or rapidly evolving ransomware variants.

Despite advancements in cybersecurity defenses, ransomware remains a dynamic and persistent threat. Attackers continually adapt their tactics to circumvent existing security measures, underscoring the need for defenses that are both proactive and adaptive. Beyond technological challenges, the economic and operational impacts of ransomware attacks necessitate a broader strategic framework that integrates legal, policy, and organizational responses. Addressing these challenges requires a multidisciplinary approach that transcends purely technical solutions.

Future research should prioritize the enhancement of real-time detection capabilities, the development of decentralized security solutions leveraging blockchain technology, and the promotion of cross-industry collaboration to facilitate the sharing of threat intelligence and best practices. Additionally, it is critical to explore the dual implications of emerging technologies, which have the potential to empower

both defenders and attackers. By examining these dimensions, researchers can better understand the evolving threat landscape and identify innovative solutions.

This study establishes a foundational understanding of ransomware's complex dynamics and its profound impact on IoT security. It emphasizes the necessity for a collective effort among researchers, practitioners, and policymakers to develop strategies that not only respond to but also anticipate the tactics of cyber adversaries. Such collaborative initiatives are essential to safeguarding the increasingly interconnected world of IoT from the escalating threat of ransomware.

CRedit authorship contribution statement

Peizhi Yan: Writing – original draft, Validation, Resources, Formal analysis, Data curation, Conceptualization. **Tala Talaei Khoei:** Writing – review & editing, Visualization, Supervision, Project administration, Conceptualization.

Declaration of competing Interest

The authors have declared no conflict of interest.

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